

4.2 Detecting equal values with branches ^

Settings

Beta features

if

An if statement executes a group of statements if an expression is

equality operator / ==

The equality operator (==) evaluates to True if the left and right side

PARTICIPATION ACTIVITY

4.2.1: Detecting if two items are equal: Hotel discount.

Start ☐ 2x speed

```
hotel_rate = 150

num_years = int(input("Enter years married: "))

if num_years == 50:
    print("Congratulations on 50 years of marriage!")
    hotel_rate = hotel_rate / 2

print(f"Your hotel rate: ${hotel_rate:.2f}")
```

```
hotel_rate = 150
num_years = int(input("Enter years married:"))

if num_years == 50:

    print("Congratulations on 50 years of marriage!")
    hotel_rate = hotel_rate / 2
```

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97	5
98	

Enter years married:
Congratulations on 50 years of marriage!
Your hotel rate: \$

↓

```
print(f"Your hotel rate: ${hotel_rate:.2f}")
```

Captions ^

1. An if statement executes a group of statements if an expression is True. The program assigns the value of the number of years the user has been married from input.
2. num_years is 50. So the expression num_years == 50 evaluates to True, and the if statement after the colon : will execute next.
3. hotel_rate is divided in half, which is the discount for guests celebrating their 50th wedding anniversary.
4. The program completes by printing the hotel rate.

inequality operator / !=

The inequality operator (!=) evaluates to True if the left and right sides are different.

Boolean

A Boolean is a type that has just two values: True or False.

Table 4.2.1: Equality and inequality operators.

Equality operators	Description	Example (assume x = 3)
==	a == b means a is equal to b.	x == 3 is True x == 4 is False
!=	a != b means a is not equal to b	x != 3 is False x != 4 is True

if-else

An if-else statement executes one group of statements when an expression is True and another group when it is False.

another group of statements when the expression is false.

PARTICIPATION ACTIVITY

4.2.5: If-else statement: Determining if a number is even or odd.

Start



2x speed

```
user_num = int(input("Enter a number: "))
div_remainder = user_num % 2

if div_remainder == 0:
    print(f"{user_num} is even.")
else:
    print(f"{user_num} is odd.")
```

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Enter a number: 22
22 is even.

Enter a number: 45
45 is odd.

Captions ^

1. An if-else statement executes a group of statements if an expression is True, and executes otherwise.
2. `user_num % 2` evaluates to the remainder of dividing `user_num` by 2. `user_num` is 22, so `div_remainder` is assigned with 0.
3. The if statement's expression `div_remainder == 0` evaluates to `0 == 0`, which is True. So the if's statements execute.
4. `user_num` is 45, so `div_remainder` is assigned with 1. The if statement's expression `div_remainder == 0` which is False. So the else's statements execute.

CHALLENGE ACTIVITY

4.2.1: Branches with equality and inequality operators: Enter the output.

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Start

Type the program's output

```
num_puppies = 4  
  
if num_puppies == 5:  
    print("c")  
else:  
    print("f")  
  
print("k")
```

f
k

1

2

Check**Next**View solution ▼ *(Instructors only)***CHALLENGE
ACTIVITY**

4.2.2: Basic if-else.

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Start

Write an if-else statement for the following:

If `barcode_check_digit` is not equal to 8, execute `group_id = 10`. Else, execute `group_id = barcode`.

Ex: If `barcode_check_digit` is 14, then `group_id = 10`.

```
1 barcode_check_digit = int(input()) # Program will be tested with values: 6
2
3 """ Your code goes here """
4
5 print(group_id)
6
```

1

Check**Next**

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elif

Additional branches use the `elif` keyword, which means "else if".

Figure 4.2.1: Multi-branch if-else statement. Only 1 branch will execute.

```

if expression1:
    # Statements that execute when expression1 is True
    # (first branch)
elif expression2:
    # Statements that execute when expression1 is False and expression2 is True
    # (second branch)
else:
    # Statements that execute when expression1 is False and expression2 is False
    # (third branch)

```

Figure 4.2.2: Multi-branch if-else example: Anniversaries.

```

num_years = int(input("Enter number years married: "))

if num_years == 1:
    print("Your first year -- great!")
elif num_years == 10:
    print("A whole decade -- impressive.")
elif num_years == 25:
    print("Your silver anniversary -- enjoy.")
elif num_years == 50:
    print("Your golden anniversary -- amazing.")
else:
    print("Nothing special.")

```

```

Enter number years marr
A whole decade -- impre
...

Enter number years marr
Your silver anniversary
...

Enter number years marr
Nothing special.
...

Enter number years marr
Your first year -- grea

```

**CHALLENGE
ACTIVITY**

4.2.3: Detect specific values.

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Start

If dimensions_number is 3, output "Three-dimensional space". Otherwise, output "Not a three-d

[Click here for examples](#) ▼

```
1 dimensions_number = int(input())
2
3 """ Your code goes here """
4
```

1

2

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